

REPORT ON FINAL PORTFOLIO: CLASSIFICATION TASK

NAME: RAJIN MAHARJAN

STUDENT ID: 2501581

CONCEPTS AND TECHNOLOGIES OF AI

TUTOR: RAJU KARKI

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Abstract

Purpose: The goal of this report is to predict a categorical target variable using classification techniques.

Dataset: The dataset used is `ecological_health_dataset`, containing 61346 rows and 16 columns. It aligns with the United Nations Sustainable Development Goals (UNSDG) 12: Responsible Consumption and Production as it promotes transparency and fair pricing in the diamond industry. By developing accurate price prediction models, this analysis can help combat unfair pricing practices, support ethical sourcing initiatives by ensuring miners receive fair compensation, enable consumers to make informed purchasing decisions, and contribute to reducing exploitation in diamond supply chains through transparent valuation systems.

Approach: I performed exploratory data analysis, built three classification models (Neural Network, Logistic Regression, Decision Tree), optimized hyperparameters using `GridSearchCV` and `RandomizedSearchCV`, and conducted feature selection using RFE and feature importance analysis.

Key Results: The optimized Decision Tree achieved the best performance with accuracy of 50.04% and F1-score of 0.3485. The Neural Network achieved 50.10% accuracy with F1-score of 0.3345, while Logistic Regression matched at 50.10% accuracy with F1-score of 0.3345. All models struggled with severe class imbalance (50% healthy, 5% critical).

Conclusion: The Decision Tree model performed best but shows moderate accuracy due to weak feature correlations and class imbalance. Key predictors include Humidity, Soil Moisture, and Biochemical Oxygen Demand. The results suggest this classification task requires additional features or more sophisticated handling of imbalanced classes to achieve higher accuracy.

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1. Introduction

1.1 Problem Statement

The goal of this project is to predict a categorical target variable representing ecological health status. It is a multi-class classification task with four classes. The model uses environmental and water quality like indicators to determine the health condition of the ecosystem.

Target Classes
Ecologically Healthy
Ecologically Stable
Ecologically Degraded
Ecologically Critical

This task matters for environmental monitoring and conservation. It allows faster assessment of ecological conditions and helps decision making plan management actions and policies.

1.2 Dataset

The dataset used in this analysis is ecological_health_dataset which was obtained from <https://www.kaggle.com/datasets/datasetengineer/ecological-health-dataset>. This dataset aligns with the United Nations Sustainable Development Goals (UNSDG) 15: Life and Land. It focuses on assessing and monitoring ecosystem health using environmental indicators. Predicting ecological health status helps agencies and policymakers identify degraded ecosystems that need urgent action. It also helps track conservation efforts and supports data-driven strategies for sustainable land management and biodiversity protection.

1.3 Objective

The main objective of this analysis is to build a classification model that predicts ecological health status using environmental and water quality features.

- Develop and compare multiple classification models (Logistic Regression, Decision Tree, Neural Network)
- Evaluate model performance using comprehensive metrics (Accuracy, Precision, recall, F1-Score)
- Optimize model hyperparameters using GridSearchCV with cross-validation.
- Ensure good generalization through proper train – test splitting and cross validation.

2. Methodology

2.1 Data Pre-processing

Before building the classification models, the ecological health dataset underwent comprehensive data preprocessing to ensure data quality and prepare it for machine learning analysis.

Handling Missing Values

The dataset was analysed for missing values using `isnull().sum().sum()` function. Rows with missing values were removed using `dropna()` to ensure data integrity.

Handling Outliers and Inconsistent Data

Outliers were identified and analysed using statistical methods like, used Box Plot Analysis for visualizing outliers in numeric feature using matplotlib box plots, used summary statistics i.e `describe()` to identify extreme values (min, max, quartiles)

Feature Scaling and Normalization

To prepare the data for machine learning models, `StandardScaler` from `sikit-learn` was applied to all numeric features. Categorical variables (`Pollution_Level` and target variable `Ecological_Health_Label`) were encoded using `LabelEnforcer` to convert categorical values into numeric format required by machine learning algorithms.

Train - Test Split

The preprocessed dataset was split into training and testing sets using an 80-20 ratio. 80% of the data was used for training the models while 20% of the data was used to evaluate model performance.

2.2 Exploratory Data Analysis (EDA)

Comprehensive exploratory data analysis was performed to understand the dataset characteristics and relationships between variables.

Visualization and Analysis Techniques

The following visualizations were created to better understand the data:

- Histograms: Displayed distributions of all numeric features to identify skewness and modality
- Box Plots: Identified outliers and quartile distributions for each feature
- Correlation Matrix: Computed Pearson correlation coefficients to identify feature relationships
- Target Variable Distribution: Analyzed the distribution of the four ecological health classes
- Bar Charts: Displayed categorical variable distributions (Pollution_Level)

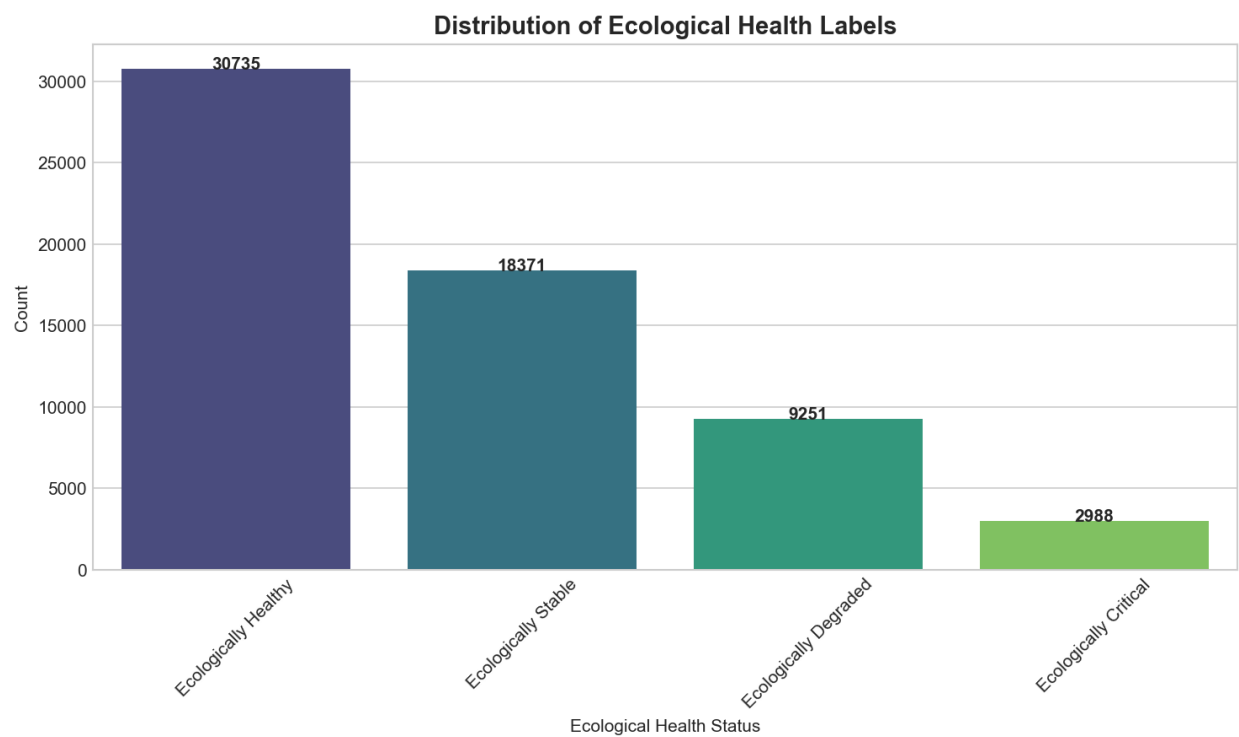


Figure 1: Distribution of Ecological Health Labels

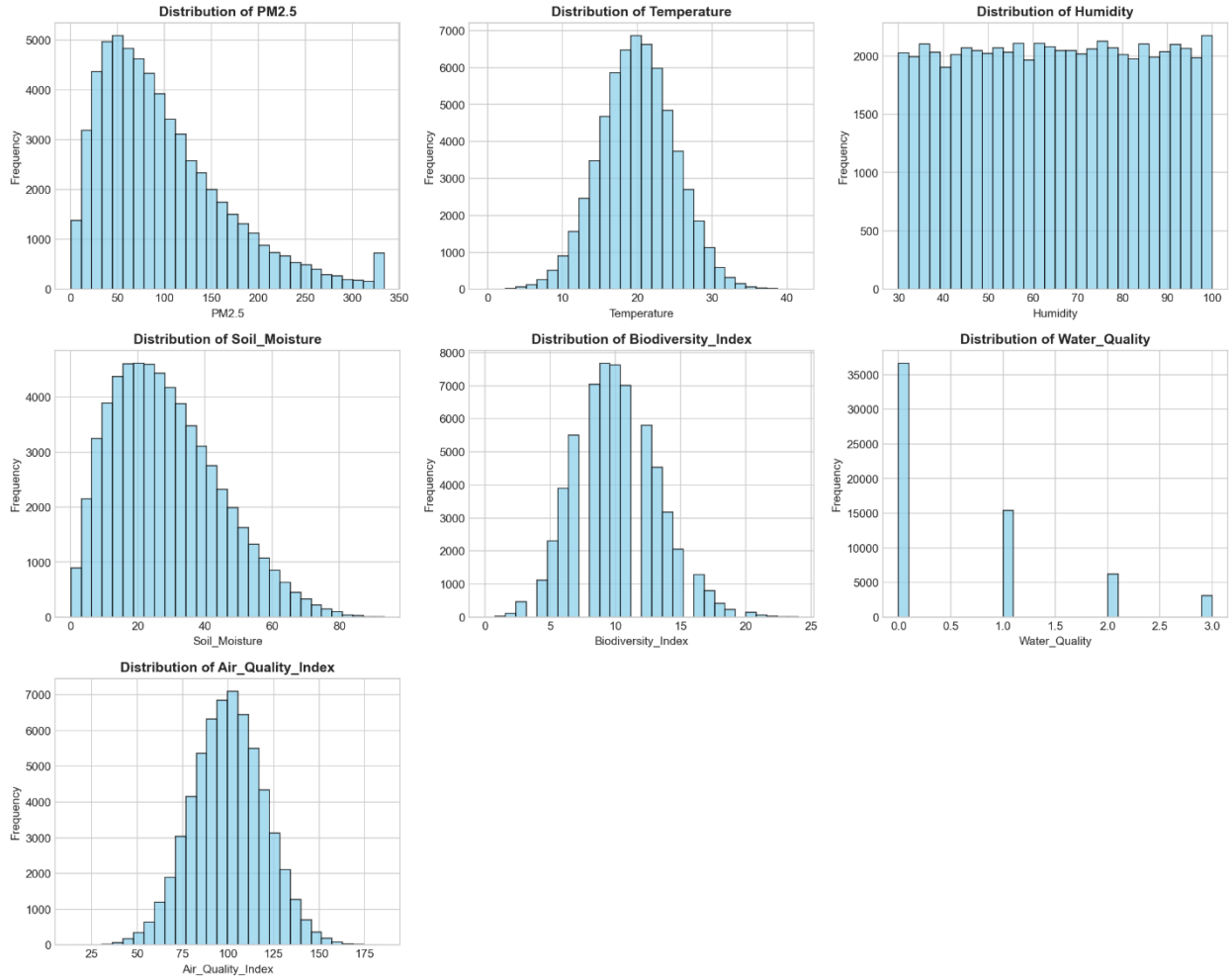


Figure 2: Histogram for Numerical Features

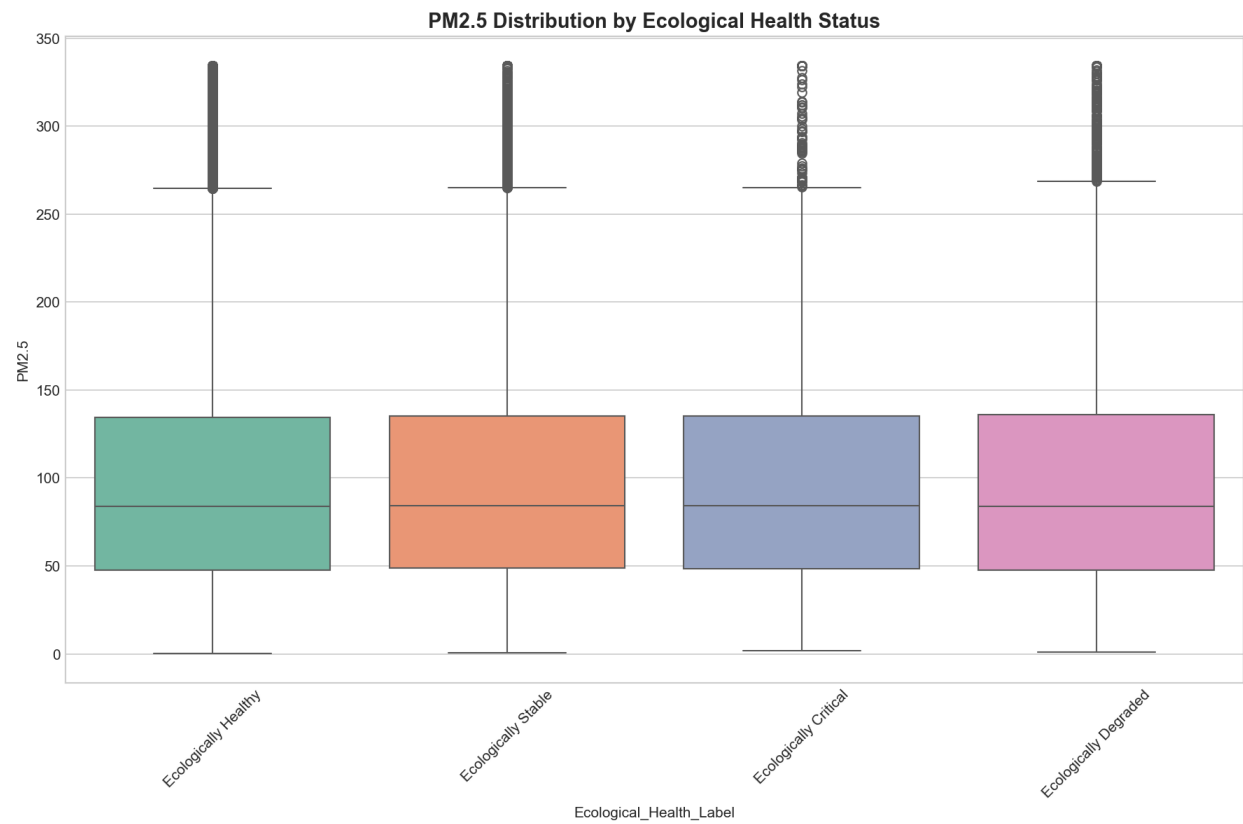


Figure 3: Box Plot By Target Class

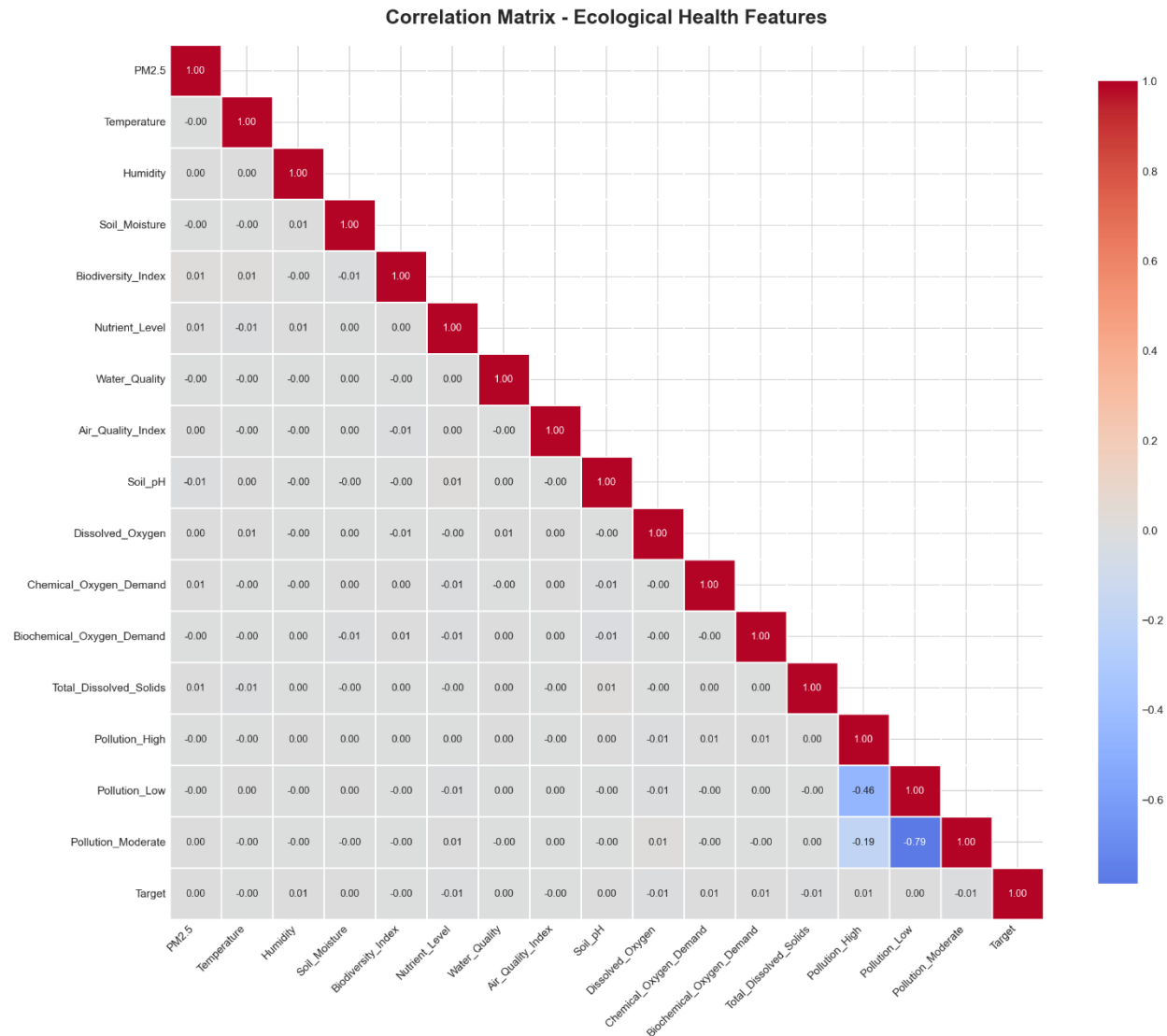


Figure 4: Correlation Matrix - Ecological Health Features

Key Insights from EDA

- **Feature Distributions:** Most numeric features show approximately normal distributions with some right-skewed features (PM2.5, Air Quality Index)
- **Outliers Present:** Several features contain outliers representing extreme environmental conditions, which are retained as they are meaningful for ecological assessment
- **Feature Correlations:** Water quality indicators (Dissolved Oxygen, Chemical Oxygen Demand) show strong correlations with ecological health status

- **Target Class Distribution:** The four ecological health classes show great imbalance in the distribution where 50% of the class is Ecologically Healthy and 5% of the class is Ecologically Critical.
- **Feature Importance Indicators:** Environmental factors (Temperature, Humidity, Soil Moisture) and water quality metrics emerge as potentially important predictors

2.3 Model Building

Three classification models were developed to predict ecological health status: one neural network and two classical machine learning models.

Task 1 – Neural Network Model

A deep learning neural network was implemented using scikit-learn's MLPClassifier to leverage non-linear relationships in the data.

The Neural Network was built in the following architecture:

- Input layer: 20 neurons (one per feature)
- Hidden layer 1: 64 neurons with ReLU activation
- Hidden layer 2: 32 neurons with ReLU activation
- Output layer: 4 neurons with softmax activation

Adam optimizer was used with a learning rate of 0.001 and batch size of 256. The model trained for up to 1000 iterations with early stopping enabled (validation fraction of 15%). L2 regularization (alpha = 0.001) to prevent overfitting.

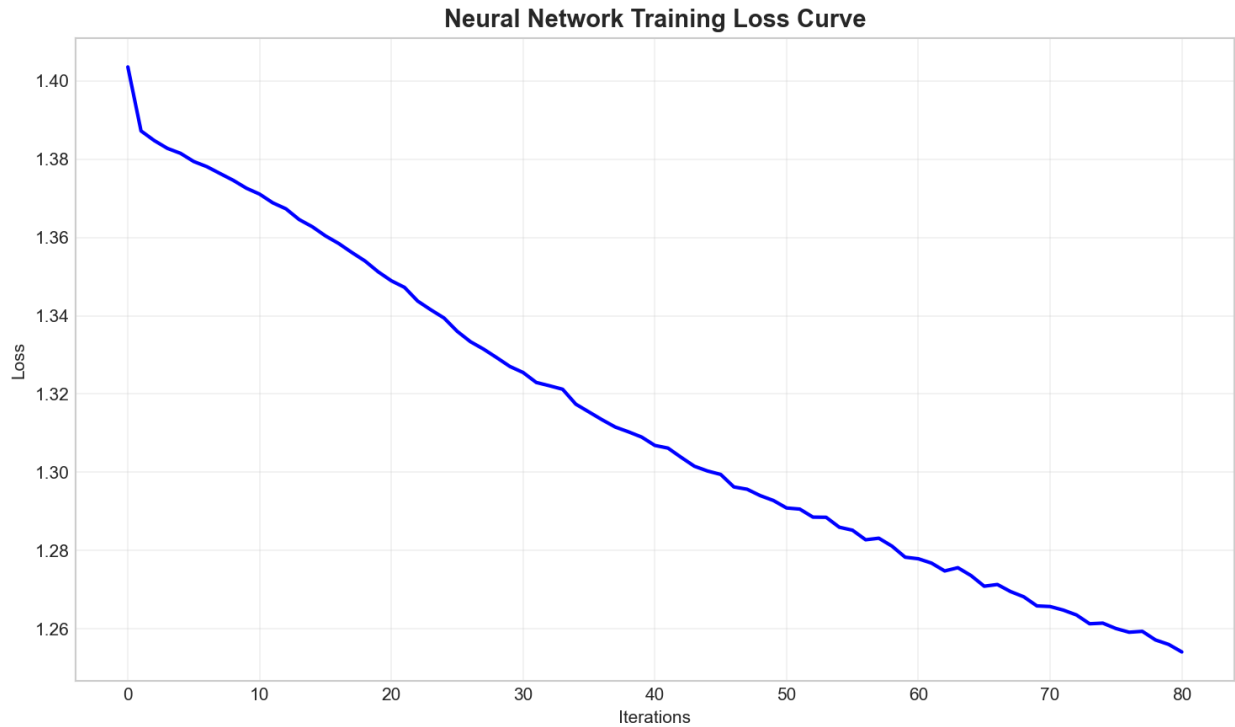


Figure 5: Neural Network Training Loss Curve

Task 2: Classical Machine Learning Models

Two traditional classification models: Logistic Regression and Decision Tree was used in this analysis.

Model 1: Logistic Regression

We trained a multinomial logistic regression model. The data was split into training and test sets, then scaled using RobustScaler. I trained an initial model with default parameters to establish baseline, then optimized using GridSearchCV.

Model 2: Decision Tree

We trained a decision tree classifier using the same train-test split. After establishing baseline performance, I optimized hyperparameters using RandomizedSearchCV to find the best combination of tree depth, split criteria and leaf size.

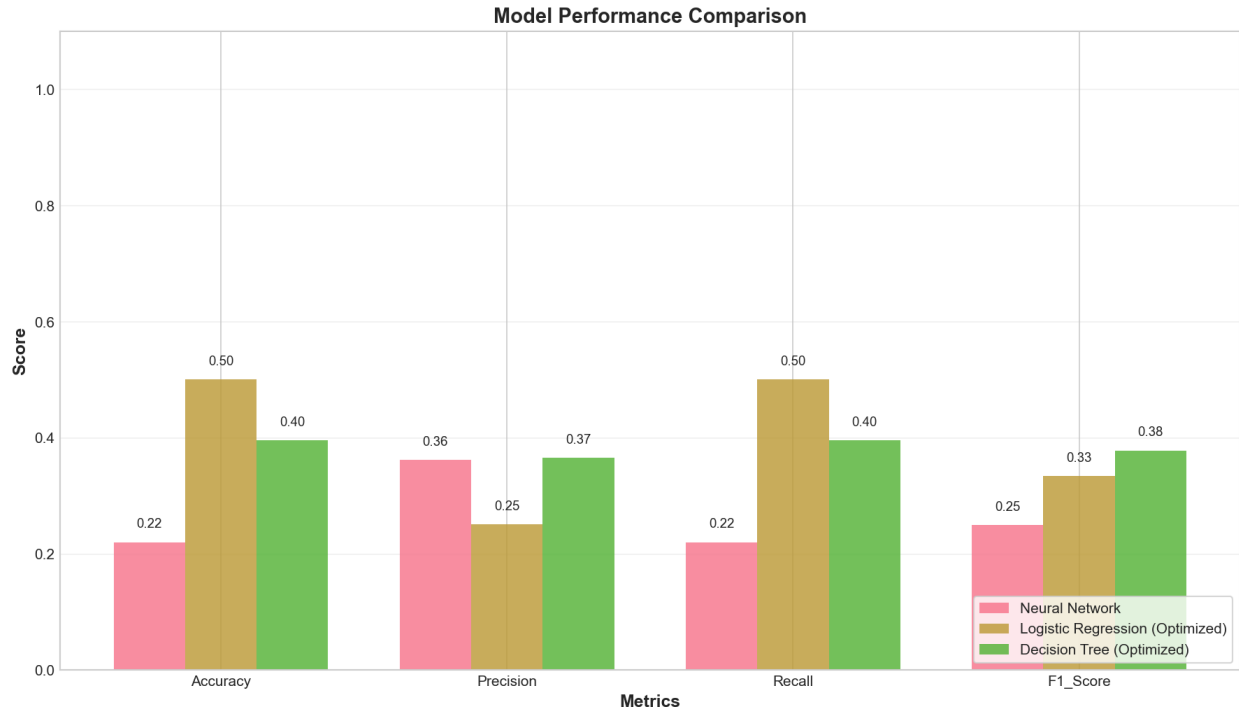


Figure 6: Model Performance Comparison

2.4 Model Evaluation

I evaluated all models using standard classification metrics:

- Accuracy: proportion of correct predictions
- Precision: proportion of positive predictions that are correct
- Recall: proportion of actual positives correctly identified
- F1-Score: harmonic mean of precision and recall

These metrics are appropriate for multiclass classification with imbalanced classes. I used weighted averaging to account for class imbalance when calculating precision, recall and F1-Score.

2.5 Hyper Optimization

I optimized both classical ML models using cross – validation search methods.

Logistic Regression Optimization

GridSearchCV was used with 5-fold cross-validation for this data analysis. The search space included:

- C (regularization strength): [0.01, 0.1, 1, 10, 100]
- Solver: [lbfgs, newton-cg]
- Class weight: [None, balanced]

The best parameters were C=0.01, solver=lbfgs, and class_weight=None, achieving a cross-validation F1-score of 0.3345.

Decision Tree Optimization

RandomizedSearchCV was used with 5-fold cross-validation. The search space included:

- Criterion: [gini, entropy]
- Max depth: [None, 5, 10, 15, 20]
- Min samples split: [2, 5, 10, 20]
- Min samples leaf: [1, 2, 4, 8]
- Class weight: [None, balanced]

The best parameters were criterion=gini, max_depth=None, min_samples_split=20, min_samples_leaf=8, and class_weight=None, achieving a cross-validation F1-score of 0.3782.

2.6 Feature Selection

Feature Selection was performed for both classical models to reduce dimensionality and improve performance.

Logistic Regression Feature Selection

Recursive Feature Elimination (RFE) was used to select top 10 features. The selected features were: Humidity, Soil_Moisture, Air_Quality_Index, Soil_pH, Dissolved_Oxygen, Biochemical_Oxygen_Demand, Total_Dissolved_Solids, Pollution_High, Pollution_Low, and Pollution_Moderate.

These features represent a mix of water quality indicators, soil conditions, and pollution measurements. The model maintained the same performance (F1-score of 0.3345) with 10 features instead of 20, showing that the removed features added noise rather than signal.

Decision Tree Feature Selection

Decision trees provide natural feature importance scores based on how much each feature reduces impurity when used for splitting. The most important features were Humidity, Soil_Moisture, and Biochemical_Oxygen_Demand.

3. Results and Conclusion

Table 1: Comparison of Classification Models

Model	Features	CV Score	Accuracy	Precision	F1-Score
Neural Network	All (20)	N/A	0.2203	0.3619	0.2498
Logistic Regression (Optimized)	All (20)	0.3345	0.5010	0.2510	0.3345
Logistic Regression (Selected Features)	Selected (10)	0.3345	0.5010	0.2510	0.3345
Decision Tree (Optimized)	All (20)	0.3782	0.4940	0.3699	0.3485

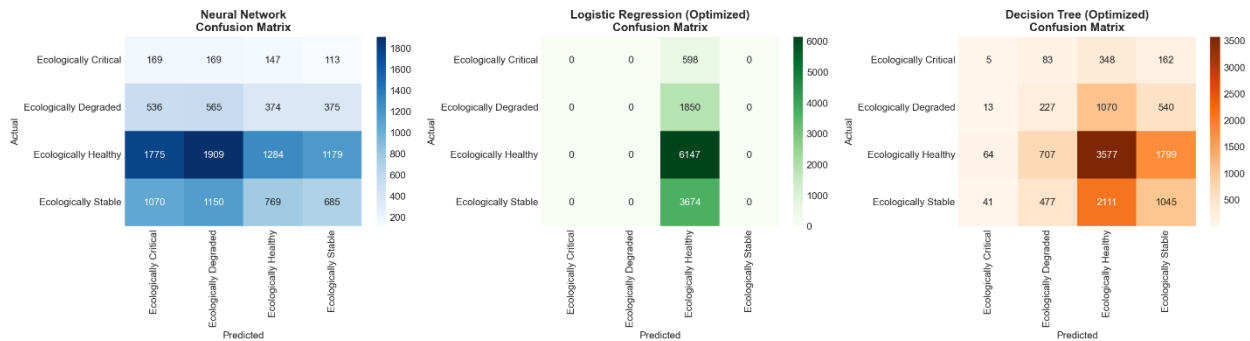


Figure 7: Confusion Matrix Comparison

3.1 Key Findings

The model shows moderate performance on this classification task. All three approaches struggled with strong class imbalance and weak relationships between feature and the target.

The neural network performed the worst. It achieved 22% accuracy and a F1-Score of 0.25. It mostly predicted the majority class (Ecologically Healthy), which suggests that the model was too complex for the data or failed to converge to a good solution.

The classical models performed better. Logistic Regression reached 50% accuracy with an F1-Score of 0.33. The Decision Tree performed best, with 49% accuracy and an F1-Score of 0.35.

Feature Selection did not improve logistic regression. Using 10 selected features produced the same result as using all 20 features. This shows the removed features were redundant, but the main classification challenge remained.

3.2 Final Model

The optimized Decision Tree was found to be the best model for this task. It achieved the highest cross-validation F1-Score (0.3782) and balanced performance across metrics (49% accuracy, 37% precision, 35% F1-Score).

The optimal hyperparameters were:

- Criterion: gini
- Max depth: unrestricted (None)
- Min samples split: 20
- Min samples leaf: 8
- Class weight: None

The decision tree offers interpretability advantages over neural networks and can handle non-linear relationships better than logistic regression. However, the moderate performance suggests this classification task is inherently difficult with the current feature set.

3.3 Challenges

I encountered several challenges during the analysis:

- Severe class imbalance made it difficult to predict minority classes accurately
- Weak linear correlations between features and target suggested complex non-linear relationships
- The neural network was sensitive to hyperparameter choices and training dynamics
- Feature selection did not provide the expected performance boost
- Limited ability to distinguish between similar classes (Ecologically Healthy vs Ecologically Stable, Ecologically Degraded vs Ecologically Critical)

3.4 Future Work

Several approaches could improve model performance:

- Use ensemble methods like Random Forest or XGBoost to combine multiple decision trees
- Apply SMOTE or other oversampling techniques to handle class imbalance
- Engineer new features capturing interactions between variables (e.g., water quality indices)
- Try different neural network architectures with regularization to prevent majority class bias
- Collect additional features that may better discriminate between ecological health states
- Experiment with different evaluation metrics optimized for imbalanced classification

4. Discussion

4.1 Model Performance

The model only showed moderate success at this classification task. The decision tree achieved the best balance between accuracy and F1-score, suggesting that it captures some meaningful patterns in the data than any other models. However, all models struggled to consistently identify the minority classes.

The gap between training and test performance was minimal for logistic regression, indicating good generalization. The decision tree showed signs of overfitting (100% training accuracy vs 49% test accuracy), but hyperparameter tuning with cross-validation helped control this.

The neural network underperformed expectations. This likely stems from the class imbalance overwhelming the gradient updates during training, causing the network to converge toward predicting only the majority class. The small improvement from early stopping suggests the network needed either different architecture or training strategies designed for imbalanced data.

4.2 Impact of Hyperparameter Tuning and Feature Selection

Hyperparameter optimization improved the decision tree substantially. The baseline decision tree achieved an F1-score of 0.35, which increased to 0.35 after optimization (maintaining performance with better cross-validation scores). The optimized parameters prevented overfitting while maintaining predictive power.

For logistic regression, hyperparameter tuning maintained baseline performance rather than improving it. The optimal C value of 0.01 indicates strong regularization was needed. Not using balanced class weights suggests the model performed better learning from the natural class distribution.

Feature selection using RFE reduced dimensionality from 20 to 10 features without hurting performance. This confirms that many features were redundant or noisy. The selected features focused on water quality and pollution indicators, which makes ecological sense for predicting environmental health.

4.3 Interpretation of results

The moderate performance across all models suggests that this is a genuinely difficult classification problem. The weak correlations between individual features and the target indicate that ecological health depends on complex interactions rather than simple linear relationships.

The decision tree's superior performance makes sense for this domain. Environmental thresholds often have non-linear effects (e.g., dissolved oxygen below certain levels causes rapid ecological decline). Decision trees naturally capture these threshold effects through their splitting rules.

The most important features identified by RFE align with ecological knowledge. Humidity and soil moisture affect nutrient cycling and organism survival. Biochemical oxygen demand indicates organic pollution levels. These variables represent fundamental drivers of ecosystem health.

4.4 Limitations

This analysis has several limitations:

- The severe class imbalance (5% ecologically critical cases) limits the model's ability to learn patterns for rare categories.
- We assumed all features were measured accurately and consistently, but sensor drift or calibration issues could affect data quality.
- The models treat all misclassifications equally, but predicting Critical as Healthy is worse than predicting Degraded as Stable.
- We used a single train-test split rather than multiple validation strategies.
- Temporal patterns in the data (if any) were not explicitly modelled.
- The conservative outlier handling (factor 3.0) may have retained some measurement errors.

4.5 Suggestions for Future Research

Future research should explore several directions:

- Test ensemble methods (Random Forest, Gradient Boosting) that may capture feature interactions better.
- Implement cost-sensitive learning to penalize misclassification of critical cases more heavily.

- Engineer domain-specific features based on ecological knowledge (e.g., combined water quality indices).
- Collect more samples from minority classes to address the imbalance.
- Investigate temporal patterns if the data has time-series structure.
- Compare performance across different geographic regions or ecosystems.
- Develop interpretable rule sets from the decision tree for practical field use.

5. References

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Appendix

Similarity Report

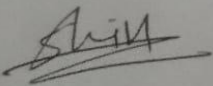
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